JIAYI MA 马佳艺

Master's candidate in Physical Education and Training, Beijing Sport University

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RESEARCH INTERESTS

Markerless motion capture and computer vision in sport · expert-aligned large language models for coaching feedback · injury-risk movement patterns in youth sabre fencing · motor skill development and analysis

EDUCATION

Beijing Sport University — Beijing, China — Sep 2024 – Jun 2027 (expected)

Master's programme in Physical Education and Training (research track), Sports Coaching College

- Thesis: *Computer vision and large language models for injury-risk assessment in youth sabre fencing*. Supervisor: Prof. Liqing Zhang
- Coursework: principles of sports training, advanced statistics, experimental research methods, advanced sport psychology, sports medicine

Springfield College — Springfield, MA, USA — Sep 2025 – May 2026

Visiting Scholar, funded by the China Scholarship Council (CSC)

- One year of full-time graduate study in an individualised coursework programme: Big Data in Sport Science; Motor Skill Development and Analysis; Evidence-Based Practice (all A grades). Passed MTEL Physical Education (22).

The Chinese University of Hong Kong — Hong Kong, China — Jun 2023

Short-term exchange, Department of Sports Science and Physical Education — cycling (A), karate (A), volleyball (A−)

Beijing Sport University — Beijing, China — Sep 2020 – Jun 2024

Bachelor of Education in Sports Training, Sports Coaching College

- Ranked 2 of 186 in cohort; Beijing Municipal Outstanding Graduate (2024). Coursework included sports medicine, sports training theory, sport psychology, and PE curriculum and pedagogy.

RESEARCH EXPERIENCE

Master's thesis — Beijing Sport University — 2024 – present

- Five-stage pipeline for sabre fencers aged 12–18: YOLOv8-Pose keypoints from dual-camera 2D video → 15 biomechanical indicators (en garde stance, sagittal- and frontal-plane lunge kinematics, inter-segment timing) → supervised fine-tuning (SFT) → preference alignment (DPO) of Qwen2.5-7B with LoRA → interpretable injury-risk reports.
- The language model is aligned to expert coaches' judgements. Ethics approval obtained from Beijing Sport University; evaluation against independent expert-coach ratings is in progress.

Haoli (毫厘) · Proof of Call — independent project — Aug 2026 – present

- Evidence-chain system for sabre right-of-way calls: aligns skeletal time series with broadcast footage frame by frame and links each call to the relevant FIE rule, making calls auditable rather than replacing referees.
- Evaluated on contested calls from the Paris 2024 Olympic Games; being integrated into the Haidian District Fencing Association's official referee-training platform. Selected for the software track of the SheNicest hackathon (1,000 participants). github.com/majiayi0526/haoli-proof-of-call

Deep Court — BroadcastCode 12 hackathon (NBA × AWS × CSDN); team captain, AI runner-up — Oct 2026

- 12-hour build of an AI broadcast system: players are identified with YOLO detection, OSNet re-identification and jersey numbers checked against the official roster; an agent on Amazon Bedrock AgentCore plans graphics and commentary, voiced with Amazon Polly. Unconfirmed identities are never named, and metrics are shown side by side rather than as causes.

Other applied projects

- **Pickleball skill analysis** (PE Department, Beijing Information Science and Technology University, 2026): MediaPipe and dynamic time warping compare strokes against 8 reference templates and return scores, parameter deviations and corrective feedback. pits-v1.vercel.app
- **Expert chain-of-thought simulation for épée training** (Beijing Sport University fencing team): a senior coach's diagnostic pathway as an executable chain of thought. epee-coach-tan.vercel.app
- **Fencing Movement Recognition Mini-Program V1.0**: registered software copyright (China); model training and development.
- **Games in junior-high PE teaching**: research group member on a key project of the Chinese Society of Education's 14th Five-Year Educational Research Plan; First Prize for research outcomes (2023).

PUBLICATIONS AND MANUSCRIPTS

- Ma, J., et al. *Concurrent validity of single-camera open-source pose estimation for joint-angle measurement against marker-based motion capture: a systematic review and meta-analysis*. Manuscript in preparation.
- Ma, J. (2022). *A study of physical conditioning techniques for fencers* [in Chinese; magazine article]. 运动·休闲·大众体育 (Sport & Leisure: Mass Sports), accepted for issue 19, 2022.

SELECTED CONFERENCE PRESENTATIONS *10+ in total, 2023–2026*

1. Ma, J. (2026, April 18). *Who coaches the coach? Augmenting pedagogical decision-making through expert-aligned AI in adolescent sabre training* [Invited talk, Postgraduate Scholar]. 17th Harvard China Education Symposium (organised by HGSE students), Cambridge, MA, USA.
2. Ma, J., & Zhang, L. (2026, March 17). *Developmental sequence of saber lunge technique: A condensed summary* [Oral presentation]. ICSPAH Symposium at the SHAPE America National Convention & Expo, Kansas City, MO, USA.
3. Ma, J., Qu, R., Xu, L., Yi, X., Gao, Y., & Zhang, L. (2025, November 9). *Markerless motion-capture-based early warning of lower-limb injuries in sabre fencers* [Oral presentation, in Chinese]. 14th National Congress on Sports Science, China.
4. Ma, J., & Zhang, L. (2025, June). *Deep-learning-based precision recognition and analysis system for sabre techniques* [Oral presentation, in Chinese]. 2nd National Conference on Artificial Intelligence in Sports, China.
5. Ma, J. (2025). *Performance training games improve fencing techniques by effecting on physical and cognitive function in high school student-fencers* [Oral presentation]. 6th Asia Sport Science Conference, Asian Society for Sport Convergence Sciences.
6. Ma, J., Qu, R., Xu, L., … Zhang, L. (2024, December). *Deep learning-based accurate recognition and analysis of saber movements* [Oral presentation]. International Conference on Sports Technology and Performance Analysis (ICSTPA 2024), Guangzhou; EI-indexed.
7. Ma, J. (2024, April). *Effects of experimental game-based training on promoting physical fitness of high-school fencing athletes* [Poster]. International Symposium on Embodied Cognition, Yale Center for Brain and Mind Health, New Haven, CT, USA.
8. Ma, J., Xu, L., Wang, S., & Sun, M. (2023, July). *An experimental study of game-based training on fencing-specific physical qualities in early school-age children* [Oral presentation, in Chinese]. 2023 National Symposium on Sports Training Science, China Sport Science Society.
9. Ma, J. (2023). *Pre-competition training of elite female discus thrower Feng Bin for the Oregon World Championships* [Plenary presentation, in Chinese]. 33rd China Collegiate Track and Field Research Conference, Federation of University Sports of China.

COACHING AND OFFICIATING

Head Coach, Sabre — The High School Affiliated to Peking University, Beijing — Sep 2020 – present

- Head sabre coach for the primary, junior-high and senior-high divisions, about 20 student-athletes per cohort over five academic years; coached athletes to 4 gold and 2 silver medals at national level and 5 gold and 3 silver medals at Beijing municipal level.
- Co-authored the school's first fencing curriculum (junior-high PE elective), adopted as the standard teaching material.

Fencing Referee — Chinese Fencing Association; Grade-A since March 2025 — 2020 – present

- 22 events officiated: 1 national-level, 3 Grade-A, 2 Grade-B and 16 Grade-C/D events. Outstanding Referee, 2023 National Modern Pentathlon Youth and Junior Championships. Also a certified tennis referee (Chinese Tennis Association, 2025).

ATHLETIC RECORD

- National Class-I Athlete, fencing (sabre)
- 2nd National Youth Games, women's sabre (Group A) — runner-up, 2019
- 15th Beijing Municipal Games, women's sabre (Group A) — champion, 2018
- China Secondary School Fencing Championships, women's sabre — runner-up, 2018
- National Youth Fencing Team training camp — selected, 2018
- Beijing Sport University Fencing Games, women's sabre — champion, 2021
- Springfield College Women's Rugby (club sport) — player, 2025–26
- Peking University "Weiming Bowl" Flag Football — champion, 2025

HONOURS AND SCHOLARSHIPS

- National Scholarship (undergraduate), Ministry of Education, 2022–23
- CSC award as a visiting scholar, Springfield College, 2025–26
- Henry Scholarship, Springfield College
- Beijing Merit Student; Beijing Municipal Outstanding Graduate, 2024
- AI runner-up, BroadcastCode 12 hackathon (NBA × AWS × CSDN), 2026
- National finalist, 19th Advanced Robotics & Simulation Technology Competition, 2026
- Outstanding Trainee, Hunan Provincial Graduate Summer School on AI and Youth Fitness (Hunan University), 2026
- Outstanding Volunteer, SHAPE America National Convention & Expo, 2026

CERTIFICATIONS AND SKILLS

Officiating	Grade-A Referee, Chinese Fencing Association (2025); USA Fencing Referee (P rating, sabre); USA Boxing Official's Pathway (2026–28); tennis, track and field, and sanda referee (China)
Teaching	Passed MTEL Physical Education (22), Massachusetts; High School PE Teacher Qualification (China); NSCA Tactical Strength and Conditioning Facilitator (TSAC-F, 2026); First Aid / CPR / AED
Vision & ML	Python, PyTorch, OpenCV, YOLOv8-Pose, MediaPipe; LoRA / LlamaFactory (SFT, DPO); full-stack prototyping with AI coding assistants
Methods	R, SPSS, SQL; ICC and weighted kappa; expert-panel design; DataCamp Data Analyst Associate (2026)
Service	Volunteer, World Para Fencing Games and ICSPAH (2026); China Open tennis (2023, Gold Award); CIFTIS (2023)
Other	Mandarin (native), English (CET-6); teaches fencing, pickleball, flag football, boxing, badminton, swimming